Sanskruti Gorle, 24070521025, A1

Open

```
Name - Sanskruti Gorle
PRN - 24070521025
Batch - A1


%{
#include <stdio.h>
#include <stdlib.h>
#include "y.tab.h"
%}

%%

[0-9]+      { yylval = atoi(yytext); return NUMBER; }
"++"        { return INC; }
"--"        { return DEC; }
"%"         { return MOD; }
[+\-*/^]  { return yytext[0]; }
[ \t]       ;
\n          { return '\n'; }
.           { return yytext[0]; }

%%

int yywrap()
{
    return 1;
}
```

```
sanskruti@LAPTOP-ICU7O78D:~$ yacc -d cc9.y
cc9.y: warning: 28 shift/reduce conflicts [-Wconflicts-sr]
cc9.y: note: rerun with option '-Wcounterexamples' to generate conflict counterexamples
sanskruti@LAPTOP-ICU7O78D:~$ flex cc9.l
sanskruti@LAPTOP-ICU7O78D:~$ gcc y.tab.c lex.yy.c -o a.out
sanskruti@LAPTOP-ICU7O78D:~$ ./a.out
10 + 5 * 2
Result: 20
sanskruti@LAPTOP-ICU7O78D:~$ ./a.out
2 ^ 5
Result: 32
sanskruti@LAPTOP-ICU7O78D:~$ ./a.out
17 % 5 + 3
Result: 5
sanskruti@LAPTOP-ICU7O78D:~$ ./a.out
++7
Result: 8
sanskruti@LAPTOP-ICU7O78D:~$ ./a.out
10+
Error: syntax error
Invalid expression
sanskruti@LAPTOP-ICU7O78D:~$ ./a.out
7 / 0
Error: Division by zero
Result: 7
sanskruti@LAPTOP-ICU7O78D:~$
```

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%{
#include <stdio.h>
#include <stdlib.h>

int yylex();
void yyerror(char *);

int power(int base, int exp)
{
    int result = 1;
    int i;

    for(i = 0; i < exp; i++)
        result = result * base;

    return result;
}
%}

%token NUMBER INC DEC MOD

%left '+' '-'
%left '*' '/' MOD
%right '^'

%%

stmt: expr '\n'    { printf("Result: %d\n", $1); }
| error '\n'   { printf("Invalid expression\n"); yyerrok; }
    ;

expr: NUMBER       { $$ = $1; }
| expr '+' expr { $$ = $1 + $3; }
| expr '-' expr { $$ = $1 - $3; }
| expr '*' expr { $$ = $1 * $3; }
| expr '/' expr { if($3 == 0)
                            yyerror("Division by zero");
                        else
                            $$ = $1 / $3;
                    }
| expr MOD expr { if($3 == 0)
                            yyerror("Modulo by zero");
                        else
                            $$ = $1 % $3;
                    }
| expr '^' expr { $$ = power($1, $3); }
| '(' expr ')'  { $$ = $2; }
| '(' expr ')'  { $$ = $2; }
| INC expr      { $$ = $2 + 1; }
| DEC expr      { $$ = $2 - 1; }
| expr INC      { $$ = $1 + 1; }
| expr DEC      { $$ = $1 - 1; }
;

%%

void yyerror(char *s)
{
    printf("Error: %s\n", s);
}

int main()
{
    yyparse();
    return 0;
}
```